

## General Info:

- Solve all tasks.
- The final Exam is the **presentation** of your solutions.
- The Codes/Scripts must run without errors on the computers of the correctors.
- You should use the Software R, no other programming language.
- Final Presentations will be on June 3rd 14:00.
- Send your final code to Erin Sprünken by mail: [erin-dirk.spruenken@charite.de](mailto:erin-dirk.spruenken@charite.de) until the end of June 2nd

**You received the dataset *Exam\_data\_6\_1.csv* and *Exam\_data\_6\_2.csv*, all your tasks refer to these datasets except stated otherwise.**

## Task 1

1. Load and clean the data sets. Give an overview over the different variables analyze them descriptively.
2. Analyze the differences between the genders descriptively.

## Task 2

**The following tasks refer to:** *Exam\_data\_6\_1.csv*

Investigate the effect of the following variables on the billing amount:

1. Date of admission
2. Duration of stay
3. Medical condition

## Task 3

**The following tasks refer to:** *Exam\_data\_6\_1.csv*

1. For this task remove all observations with inconclusive test results.
2. Fit a logistic regression model with test result as the dependent outcome. You are free to choose which independent variables to include.

## Task 4

**The following tasks refer to:** *Exam\_data\_6\_2.csv*

Income analysis:

1. Is there an association between the income level and the distance to the hospital?
2. Are there any clinical variables which differ significantly between income levels?
3. How is age distributed in each income level?

## Task 5

**The following tasks refer to:** *Exam\_data\_6\_2.csv*

Readmission prediction:

1. Try to predict which patients are readmitted to the hospital.
2. Try different model specifications and discuss them.
3. Which variables are most important?